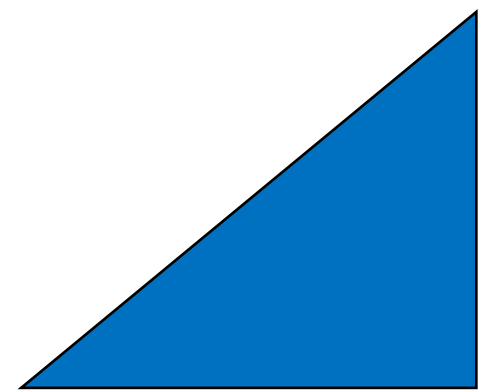




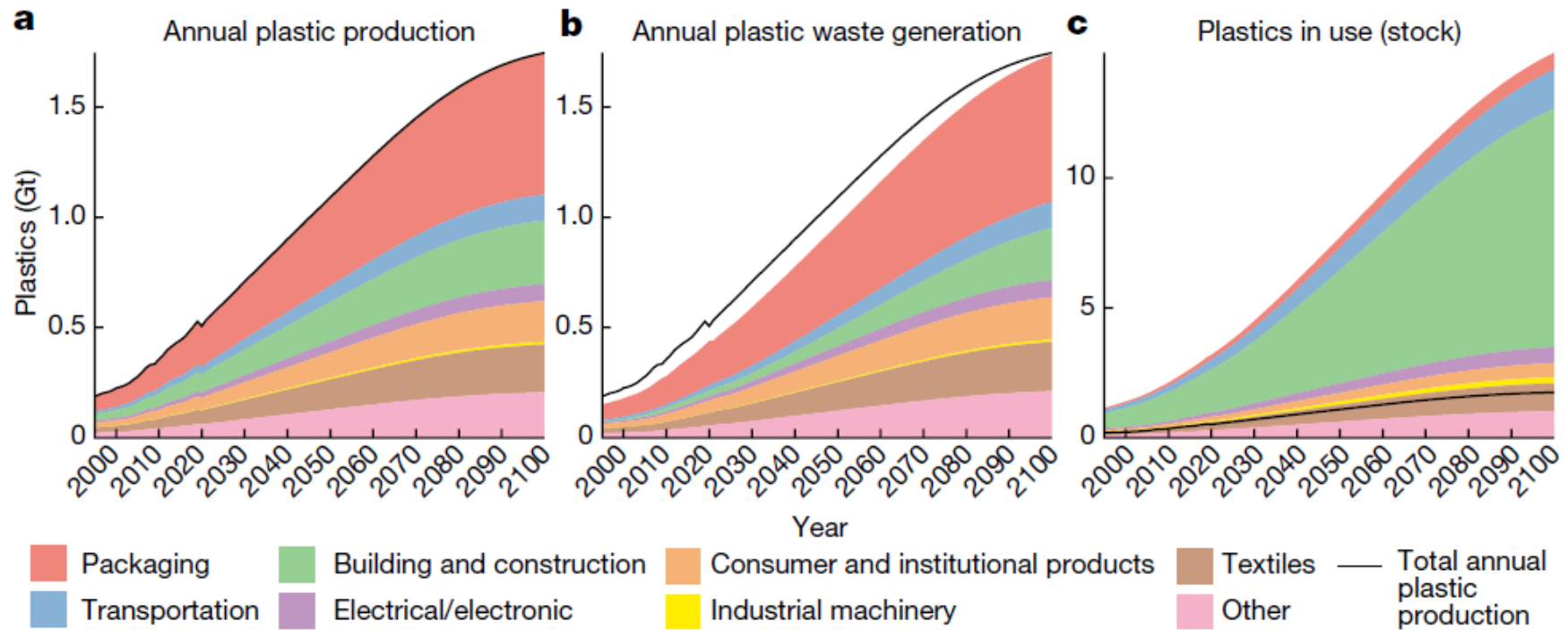
Materiais e Desenvolvimento Sustentável

Plásticos: produção e reciclagem

42295
2025/2026

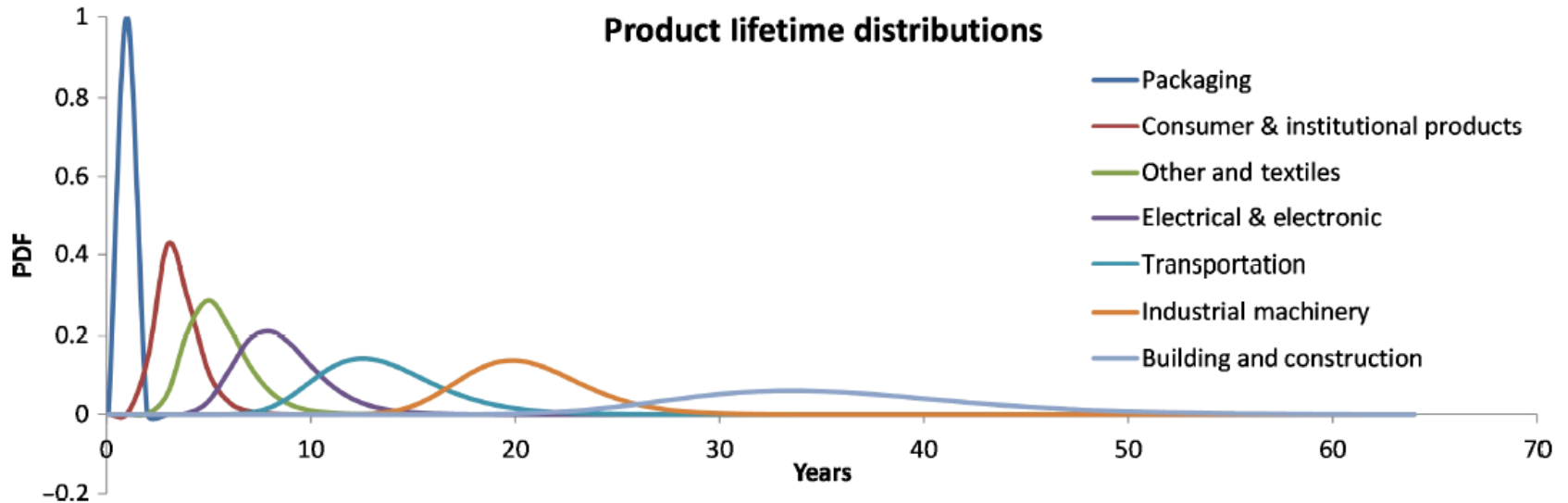


Projections of global plastic production, waste generation and plastic stocks in use, by sector



The growth in plastic demand shown in the figure implies a further increase in the GHG emissions of plastic production if there are no substantial changes in feedstock and process energy use.

Product lifetime distributions



Plastic waste generated since 1950

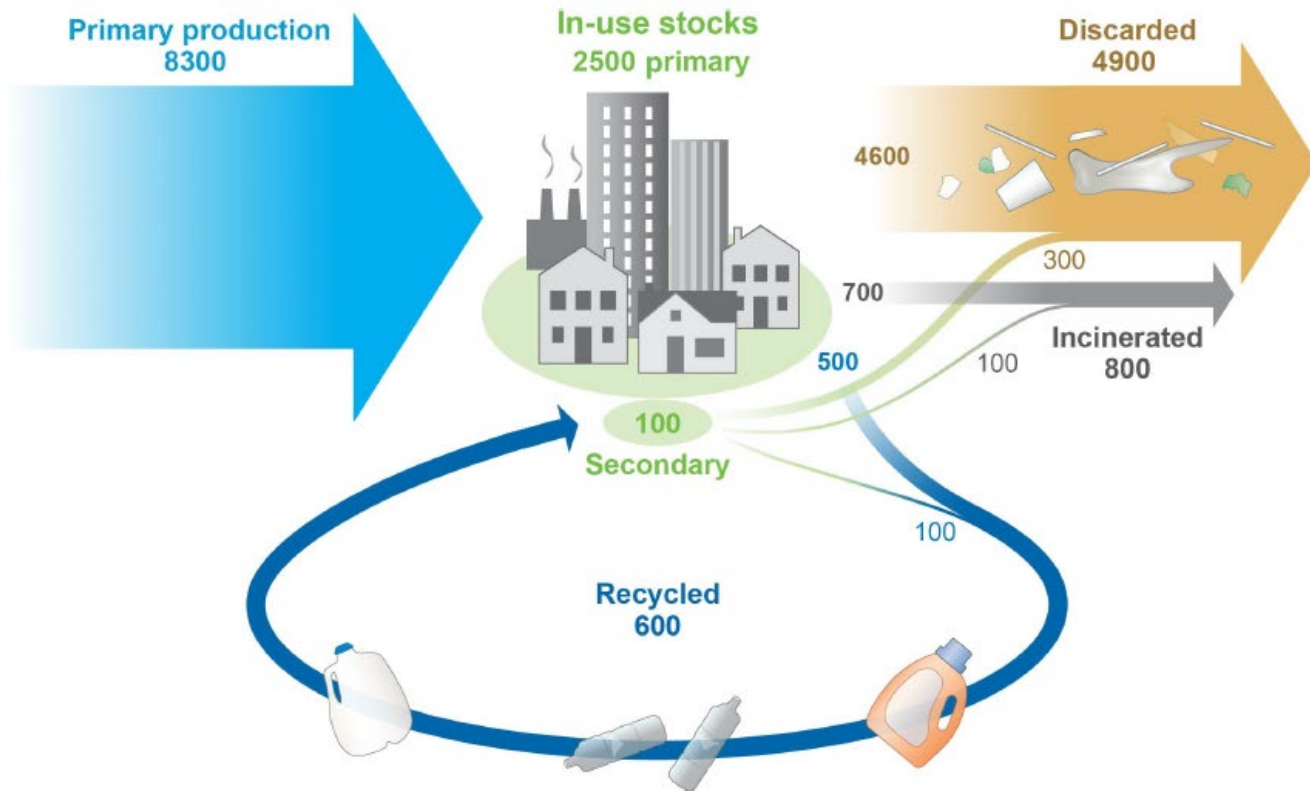


Fig. 2. Global production, use, and fate of polymer resins, synthetic fibers, and additives (1950 to 2015; in million metric tons).

“As of 2015: 6300 Mt of plastic waste had been generated.”

Environmental impact of plastic

7 of the most common* marine litter items found in coastal areas are plastic packaging



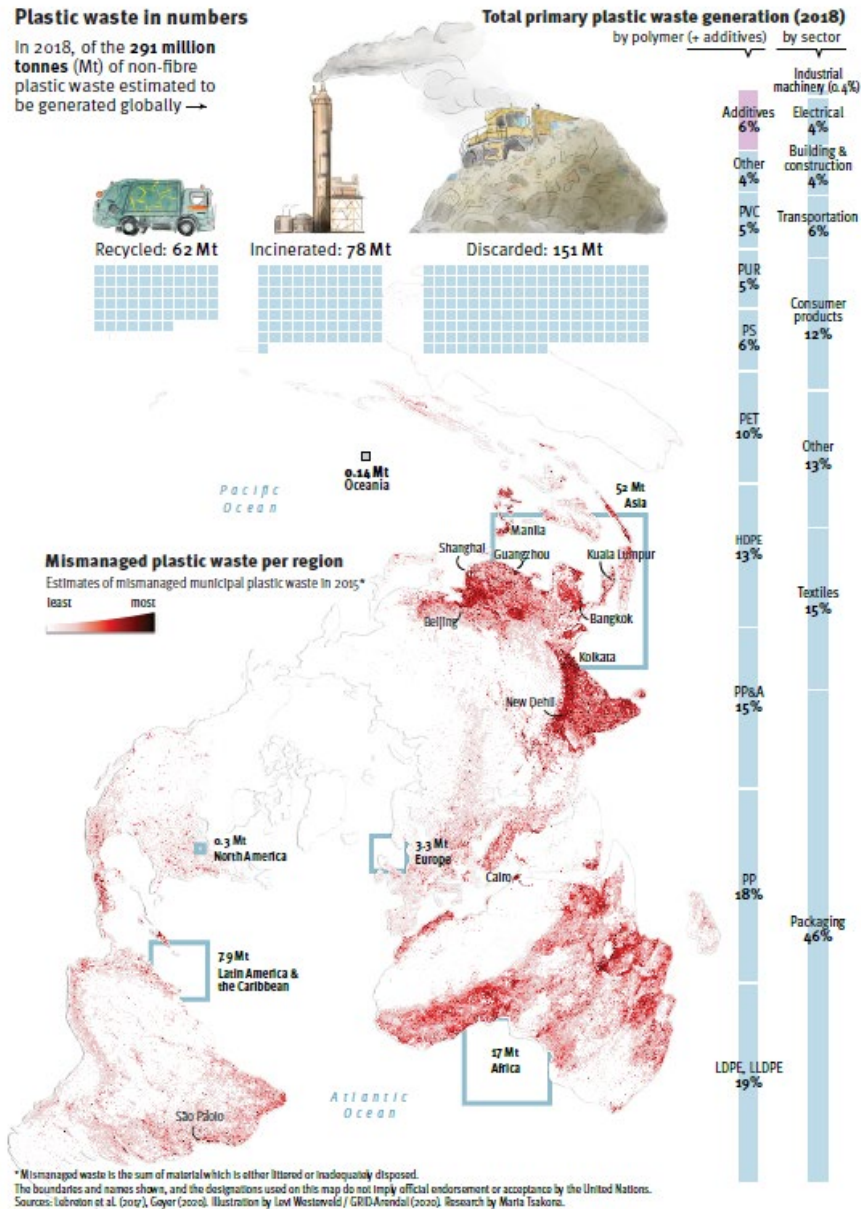
Source: “Drowning in plastics – Marine Litter and Plastic Waste Vital Graphics”, United Nations Environment Programme (2021).

Source: <https://www.europarl.europa.eu/news/en/headlines/society/20181005STO15110/plastic-in-the-ocean-the-facts-effects-and-new-eu-rules>

SINGLE-USE PLASTICS REDUCING MARINE LITTER



Environmental impact of plastic



Source: “Drowning in plastics – Marine Litter and Plastic Waste Vital Graphics”, United Nations Environment Programme (2021).

Environmental impact of plastic

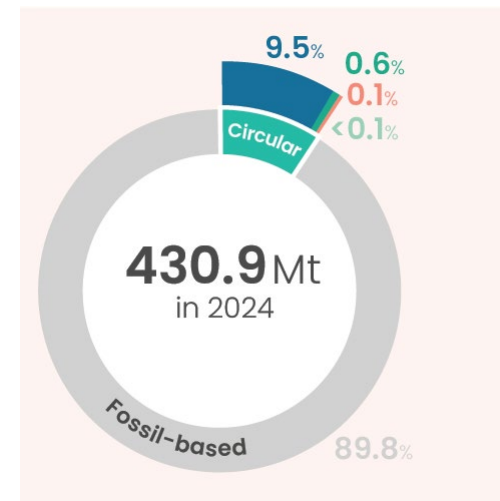
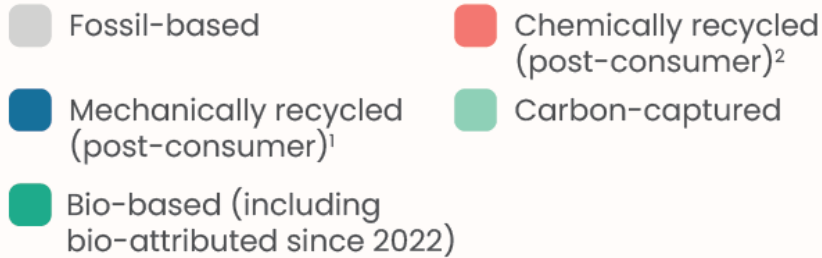
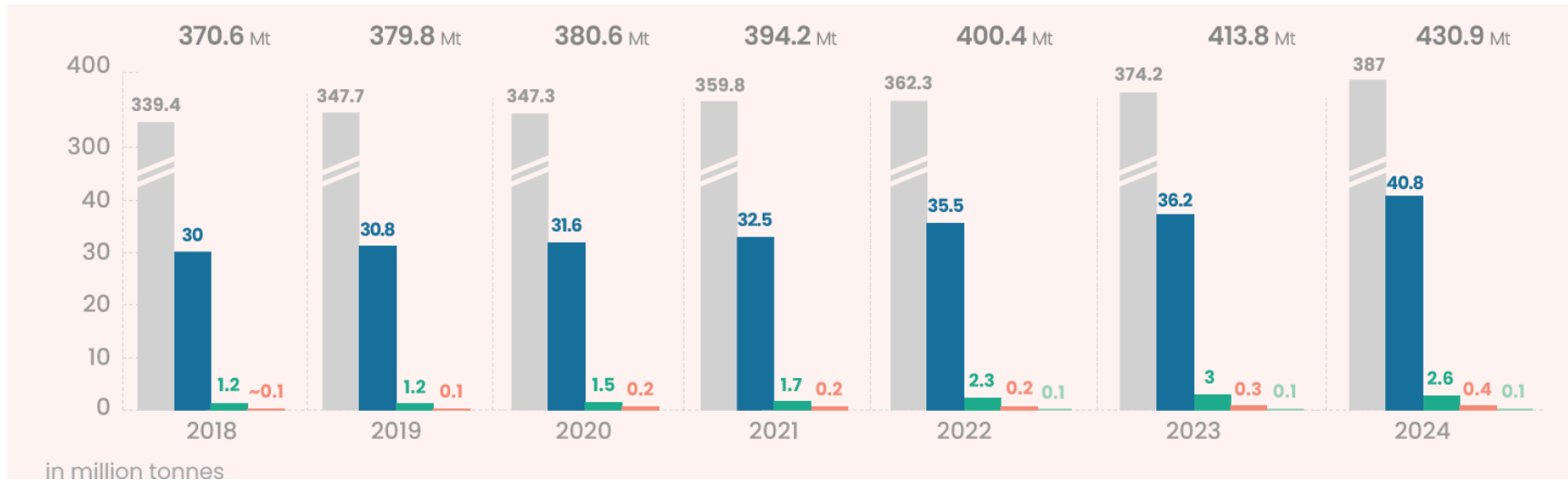
“We will not recycle our way out of the plastic pollution crisis: we need a systemic transformation to achieve the transition to a circular economy”

Inger Andersen

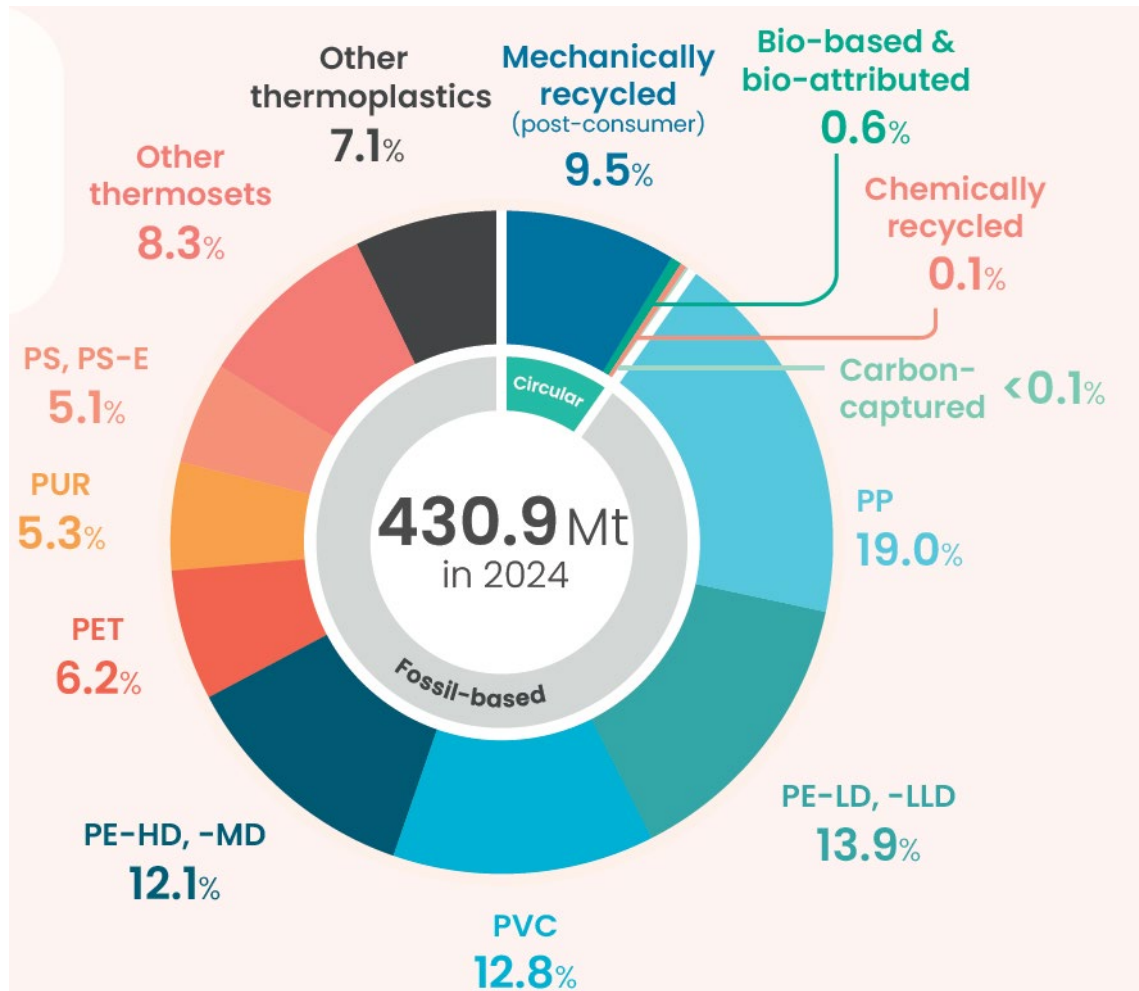
Under-Secretary-General of the United Nations and
Executive Director of the UN Environment Programme



World plastics production (2018-2024)

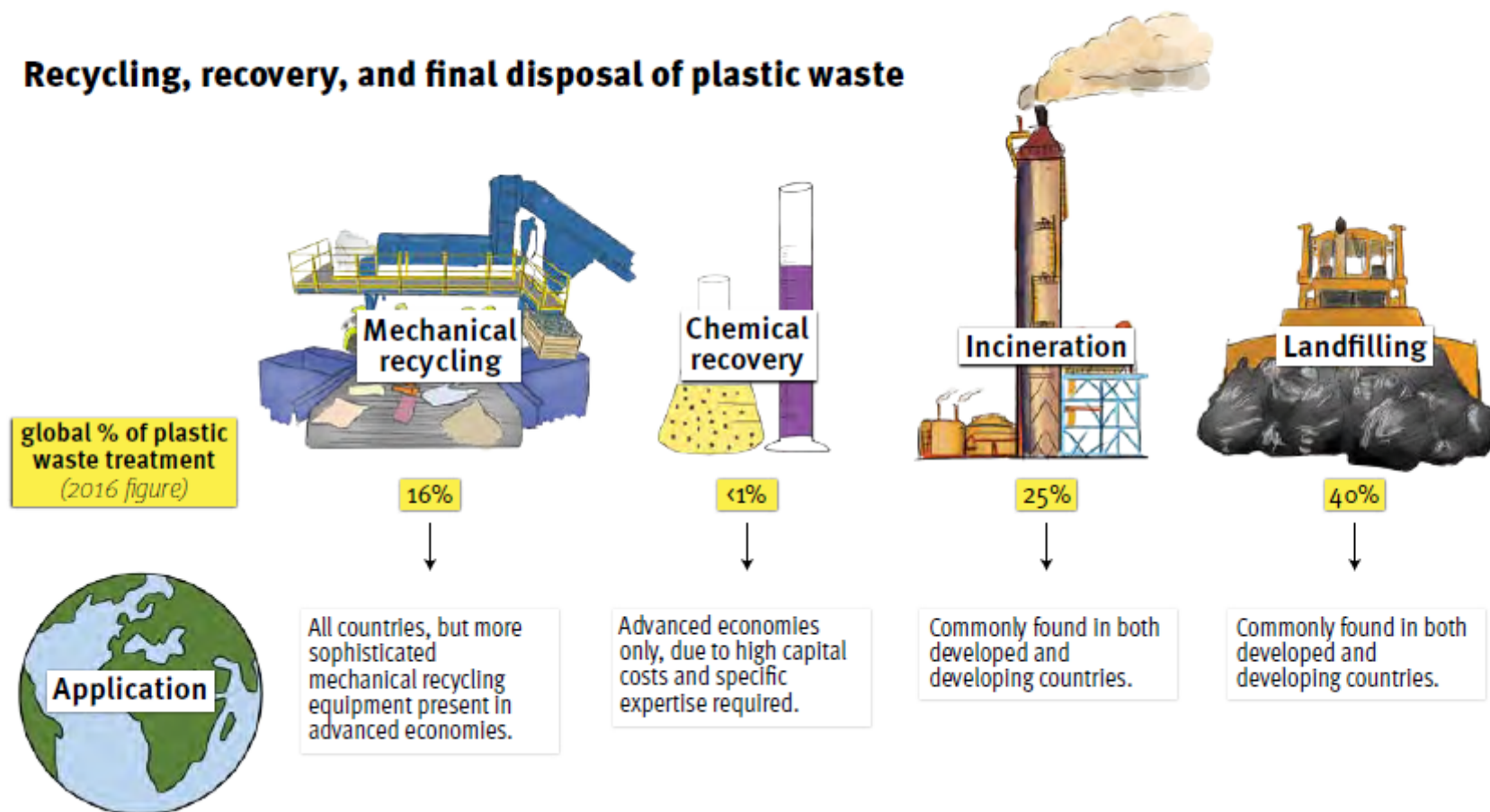


Global plastics production by polymer (2024)



Global plastic waste management rates in 2016

Recycling, recovery, and final disposal of plastic waste



Source: "Drowning in plastics – Marine Litter and Plastic Waste Vital Graphics", United Nations Environment Programme (2021).

Directive (EU) 2018/852

14.6.2018

EN

Official Journal of the European Union

L 150/141

DIRECTIVE (EU) 2018/852 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 30 May 2018

amending Directive 94/62/EC on packaging and packaging waste

(Text with EEA relevance)

THE EUROPEAN PARLIAMENT AND THE COUNCIL OF THE EUROPEAN UNION,

(g) no later than 31 December 2025
regarding the following specific materials:

- (i) 50 % of plastic;
- (ii) 25 % of wood;
- (iii) 70 % of ferrous metals;
- (iv) 50 % of aluminium;
- (v) 70 % of glass;
- (vi) 75 % of paper and cardboard;

(i) no later than 31 December 2030
regarding the following specific materials:

- (i) 55 % of plastic;
- (ii) 30 % of wood;
- (iii) 80 % of ferrous metals;
- (iv) 60 % of aluminium;
- (v) 75 % of glass;
- (vi) 85 % of paper and cardboard;

Plastics recycling rates

(packaging waste):

- 50% (2025)
- 55% (2030)

The Packaging and Packaging Waste Regulation (PPWR)



Official Journal
of the European Union

EN
L series

2025/40

22.1.2025

REGULATION (EU) 2025/40 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 19 December 2024

on packaging and packaging waste, amending Regulation (EU) 2019/1020 and Directive (EU) 2019/904, and repealing Directive 94/62/EC



Less wasteful

Plastic packaging must be made in part from recycled content, with increasing targets for 2030 and 2040.



More recyclable

All Packaging must be recyclable by 2030: This means packaging must be designed so that all parts can be used for something else afterwards, instead of valuable material being dumped or burned.



Clearly labelled

No more confusing labels or complicated colours, making it easier to sort trash for recycling.

People will be able to see **what it's made of**, **where to bin it**, how to **return it for reuse**.

The Packaging and Packaging Waste Regulation (PPWR)



Common-sense packaging

No more **layers of useless packaging or pellets** in your deliveries. Small, light packaging without empty space.



Encourages re-use, refill & collection

Deposit and return systems will be boosted.

Some single-use plastics will be banned.

Companies must **make reuse or refill options available** whenever possible, with no extra charge.



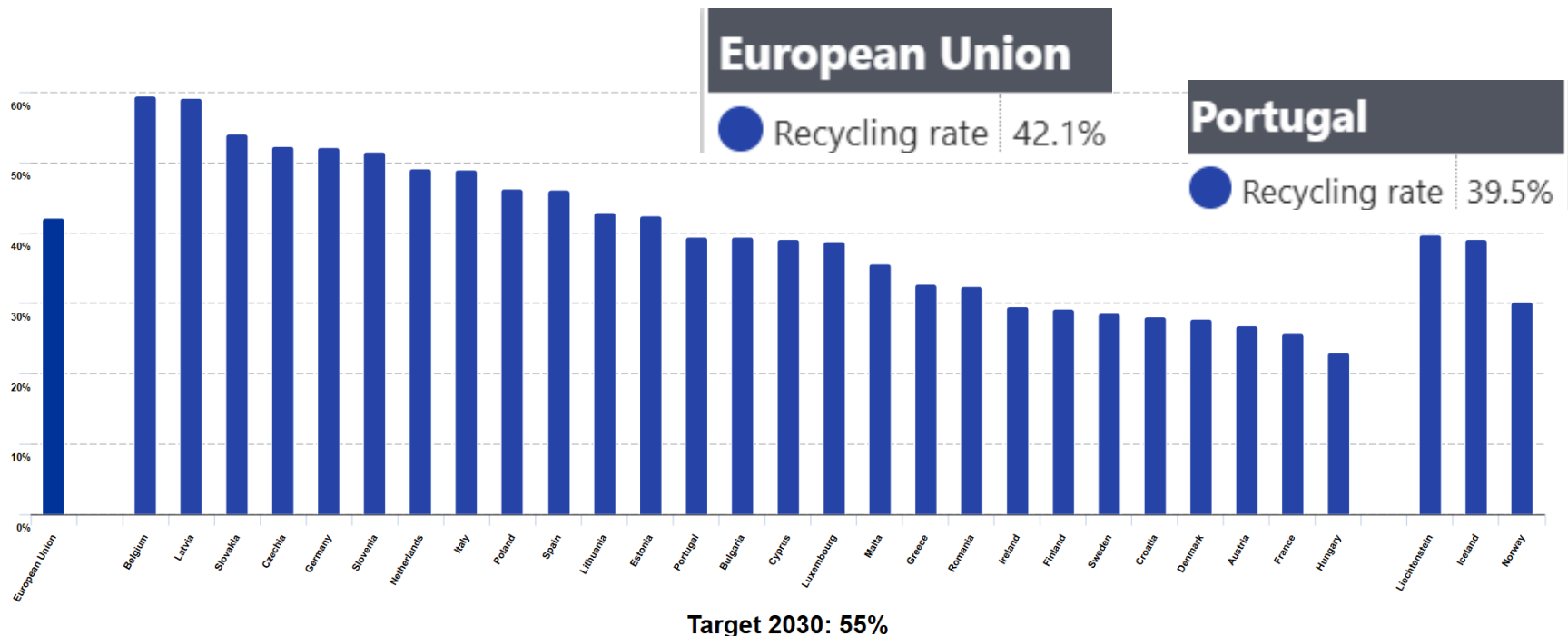
Fairer and safer

Brands using non-recyclable or environmentally harmful materials will have to **pay to clean them up**.

Harmful and cancer-causing "forever" chemicals (PFAS) will be banned from packaging from August 2026.

Recycling rate of plastic packaging waste, 2023 (EU context)

The Packaging Waste Directive sets out a target of minimum 55% of plastic packaging waste (by weight) being recycled by 2030. Belgium (59.5%) and Latvia (59.2%) were the only EU countries that achieved this target in 2023.



Plastic recycling?!

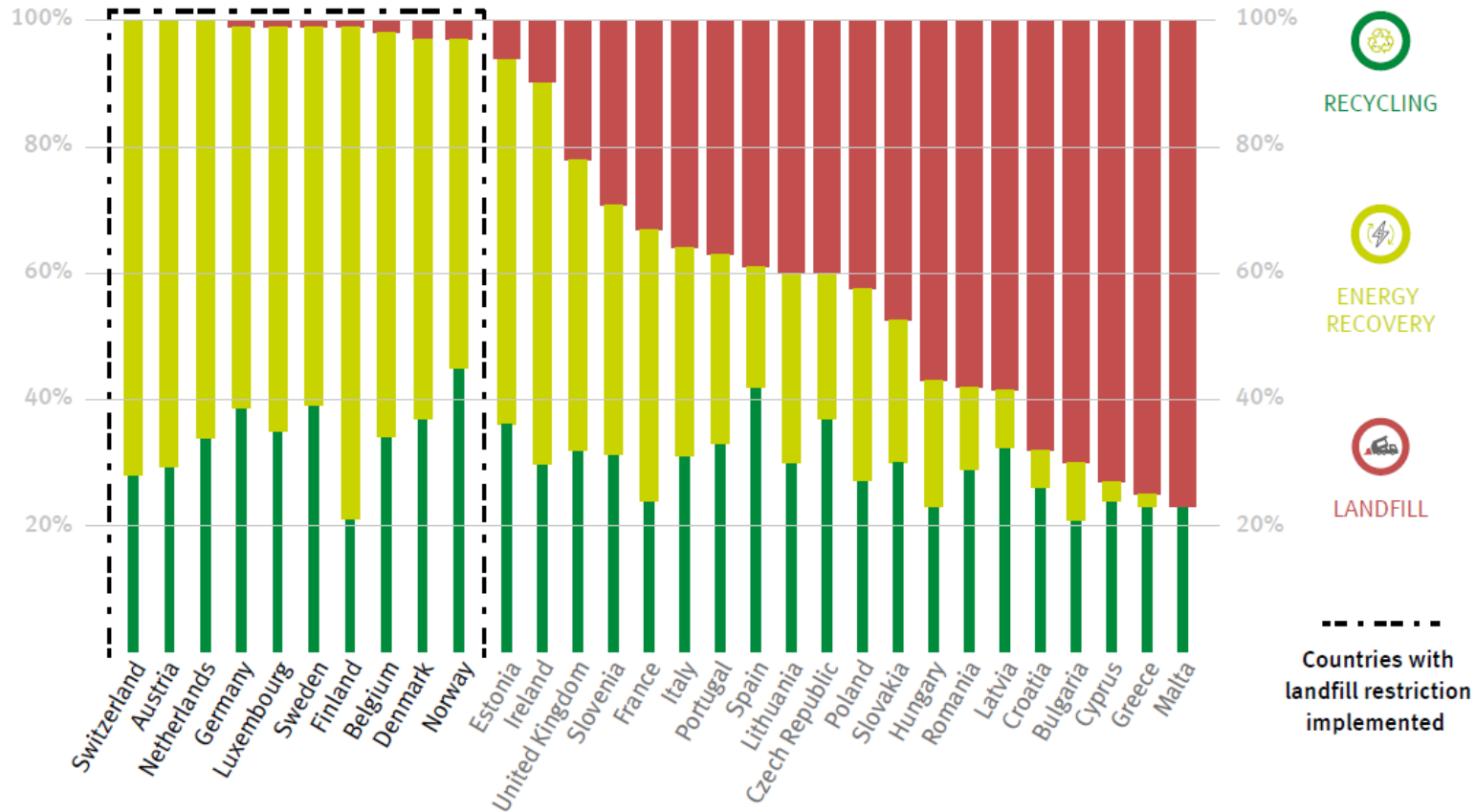


Miquel Roset, Director and Spokesperson at Retorna, states:

"The case of Spain, the home country of Vice-President designate Teresa Ribeira, is emblematic of the problem with falsified reporting on recycling results in Europe. Four months ago, we presented a report, together with Zero Waste Europe and Eunomia, to inform the Spanish government that the real figure for separated collection of plastic beverage bottles under three litters was 36%, almost half of the 71% claimed by the PRO Ecoembes. By the end of October, Spain must report whether this figure has reached 70% and, if that is not the case, the Spanish Waste Law dictates the implementation of a deposit and return system."

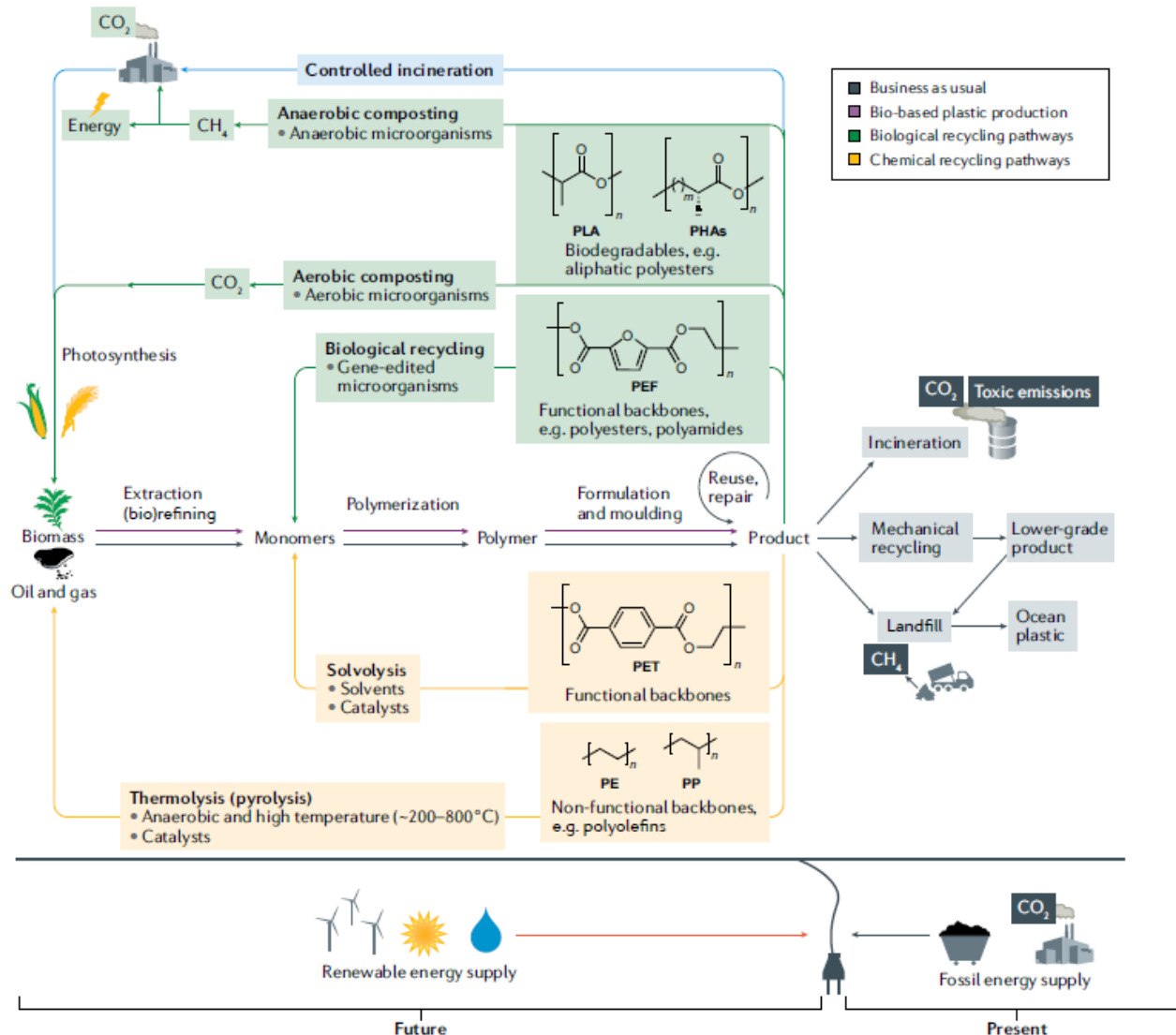
The “circular economy” of plastics (EU context)

Plastic post-consumer waste rates (in 2018 per country)



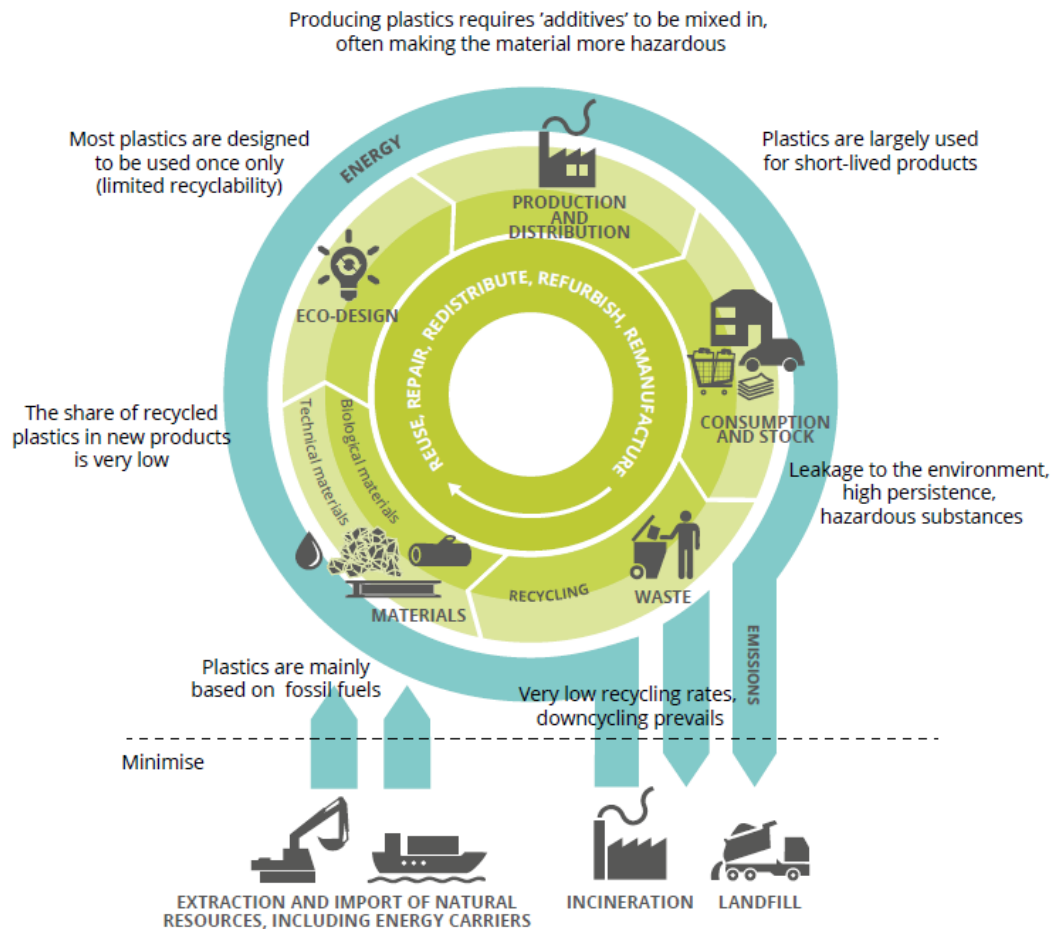
ZERO LANDFILLING IS NEEDED TO ACHIEVE
THE CIRCULAR ECONOMY OF PLASTICS!

What is the scenario when considering plastics?



Source: “Bioplastics for a circular economy”, Rosenboom, Langer, & Traverso. Nat Rev Mater 7, 117–137 (2022).

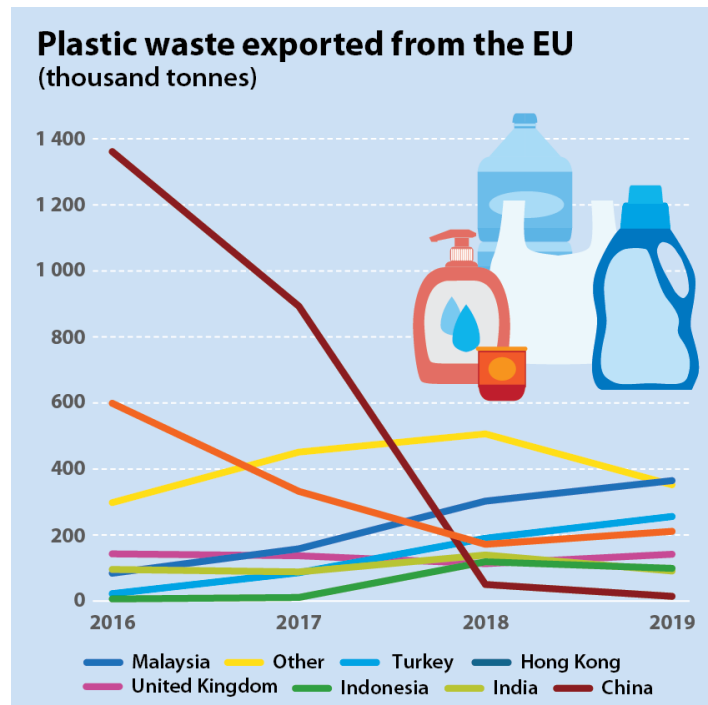
Circular economy of plastics: challenges



Plastics: measures towards future circular economy

1. Ban exports of plastic waste to third countries

Rather than shipping huge amounts of plastic waste to third countries that often do not have the necessary capacity to deal with it in a sustainable way, Europe should manage its own plastic waste. This is better from both an environmental and an ethical perspective, even if part of the waste has to be recovered for energy.



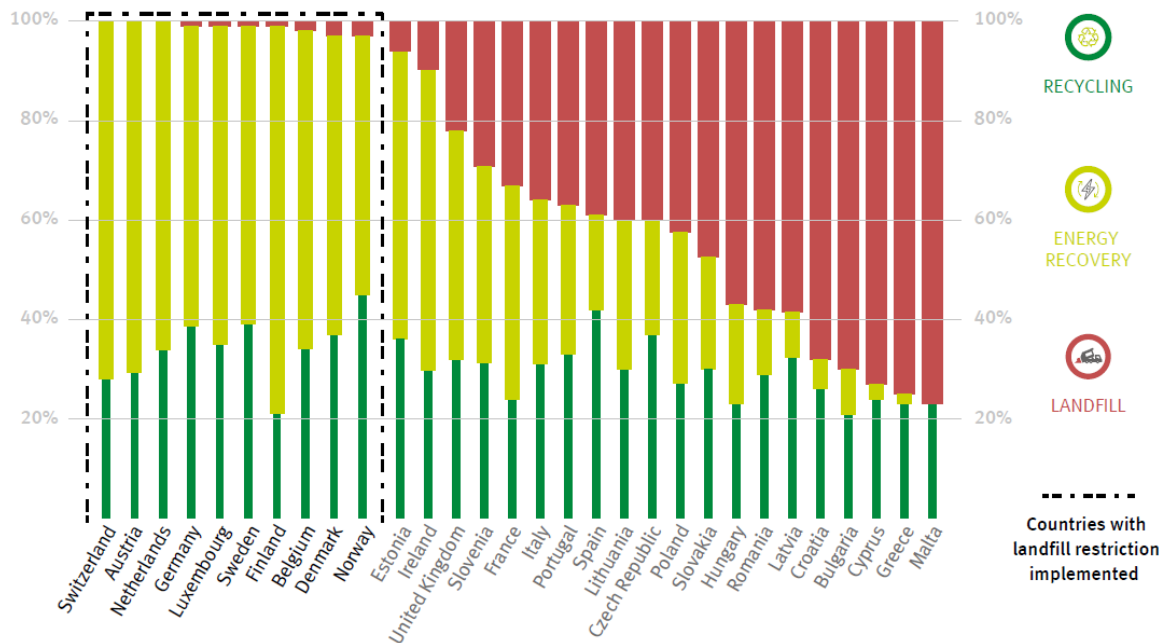
Source: “Plastics, the circular economy and Europe’s environment — A priority for action”, European Environmental Agency (2021)

Source:

<https://ec.europa.eu/eurostat/web/products-eurostat-news/-/ddn-20200709-01>

Plastics: useful measures towards future circular economy

2. Adopt a target of zero plastic waste to landfill, and minimise consumption and one-way use



Source: “Plastics- the Facts 2020”,
Report from the PlasticsEurope.

In addition to adopting a target of zero plastic waste to landfill and making reduction in consumption an explicit objective, policymakers should extend deposit refund schemes to cover a wider range of containers and single-use beverages.

Plastics: useful measures towards future circular economy

3. Extend producer responsibility (EPR)

Ambitious EPR schemes should include measures that facilitate product design choices that consider end-of-life use and environmental impacts, such as toxicity, durability, reusability, repairability and recyclability/compostability.



Deposit-refund systems and EPR schemes can increase return and collection rates for post-consumer plastics and increase the quality of the plastic collected.

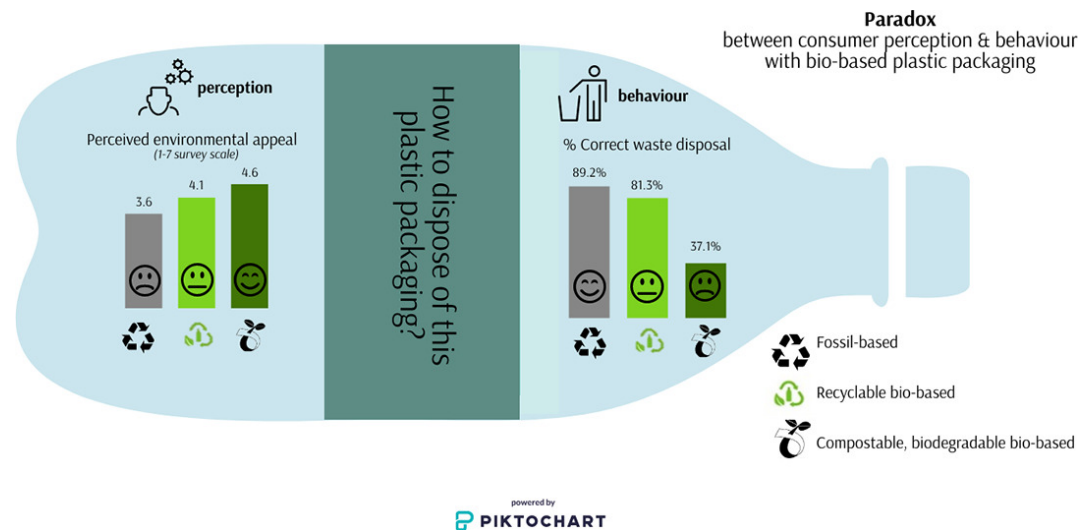
Source: “Bioplastics for a circular economy”, Rosenboom, Langer, & Traverso. Nat Rev Mater 7, 117–137 (2022).

Plastics: useful measures towards future circular economy

4. End misleading information about bio-based alternatives

“consumers may be misled by the diversity of existing labelling schemes and are often not aware of the environmental impacts associated with bio-based alternatives. A uniform European labelling scheme that relates to the actual rather than theoretical recyclability of bio-based plastics should therefore be created.”

Source: “Plastics, the circular economy and Europe’s environment — A priority for action”, European Environmental Agency (2021)



Source: “The paradox between the environmental appeal of bio-based plastic packaging for consumers and their disposal behaviour”, Taufik *et al.*, *Science of the Total Environment* 705 (2020) 135820.

Plastics: useful measures towards future circular economy

5. Advanced recycling and reprocessing technology

To extract more value from plastic waste, advanced recycling and reprocessing technology must be developed. In addition, recycling for use in the same product (closed-loop recycling) must be prioritised over other options, such as recycling for use in the production of different products (open-loop recycling) or energy recovery.

Source: “Plastics, the circular economy and Europe’s environment — A priority for action”, European Environmental Agency (2021)

Plastics: useful measures towards future circular economy

6. Limit additives and types of resin to improve recyclability

To increase the recyclability of plastics, the use of additives must be reduced and the number of polymers that can be used for specific products simplified.

Source: “Plastics, the circular economy and Europe’s environment — A priority for action”, European Environmental Agency (2021)

Five types of plastic additives



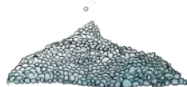
Functional

Includes, for example, stabilizers, antistatic agents, flame retardants, plasticizers, lubricants, slip agents, curing agents.



Colourants

Substances such as dyes or pigments added to give colour to plastic. Some of them are added to give a bright transparent colour.



Fillers

Added to change and improve physical properties of plastics. They can be minerals, metals, ceramics, bio-based, gases, liquids, or even other polymers.



Reinforcement

Used to reinforce or improve tensile strength, flexural strength and stiffness of the material. For example: glass fibres, carbon fibres.

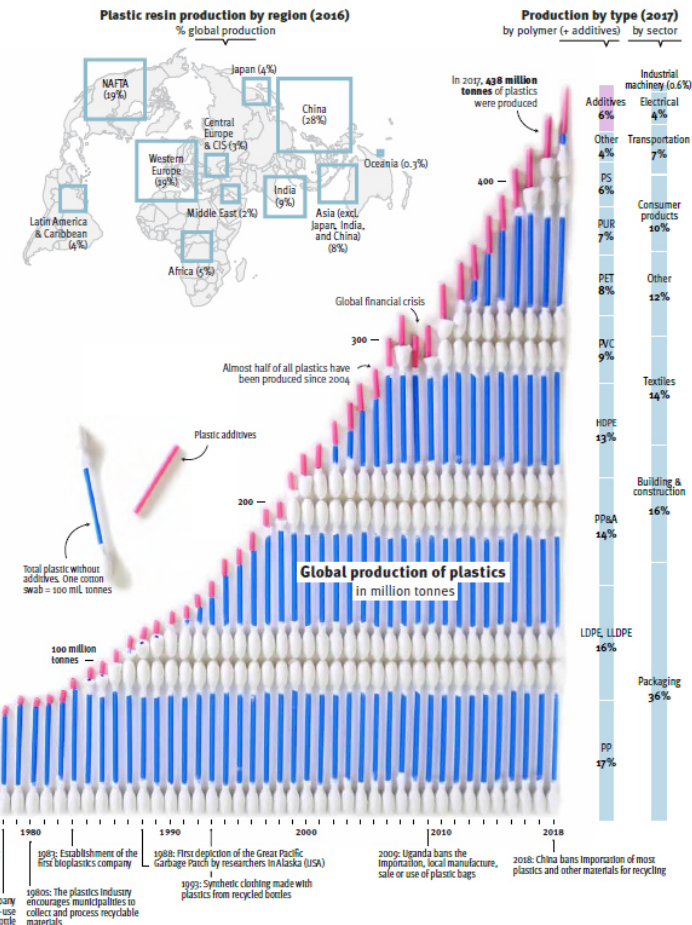


NIAS

Non-Intentionally added substances. They arrive in products from processes, such as reaction by-products or breakdown products.

Source: Hansen et al. (2013). Illustration by GRID-Arendal (2020).

Images taken and adapted from: “Drowning in plastics – Marine Litter and Plastic Waste Vital Graphics”, United Nations Environment Programme (2021).



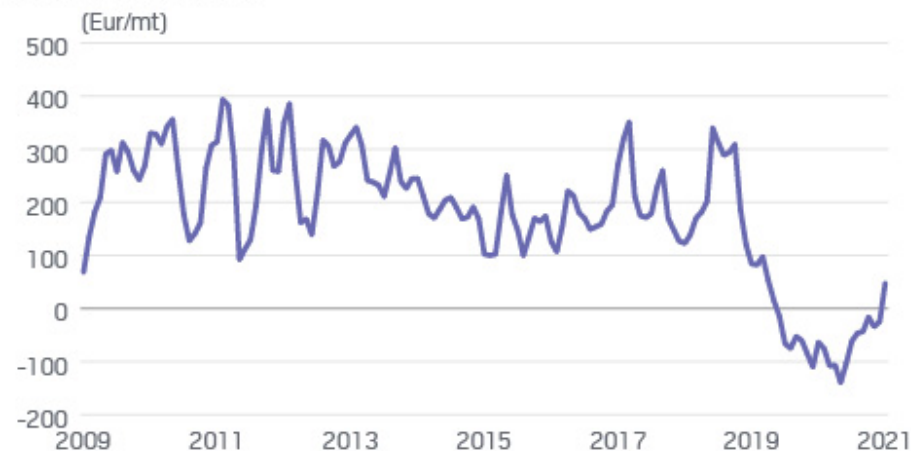
Plastics: useful measures towards future circular economy

7. Price regulations and quotas for recycled content

The current cost of virgin plastic feedstock is very low and does not include costs to the environment and climate. Policymakers should therefore adopt a regulatory and financial framework, including, for instance, a plastics tax or a requirement for minimum recycled contents, that takes into account adverse impacts across the plastic product life cycle.

Source: “Plastics, the circular economy and Europe’s environment — A priority for action”, European Environmental Agency (2021)

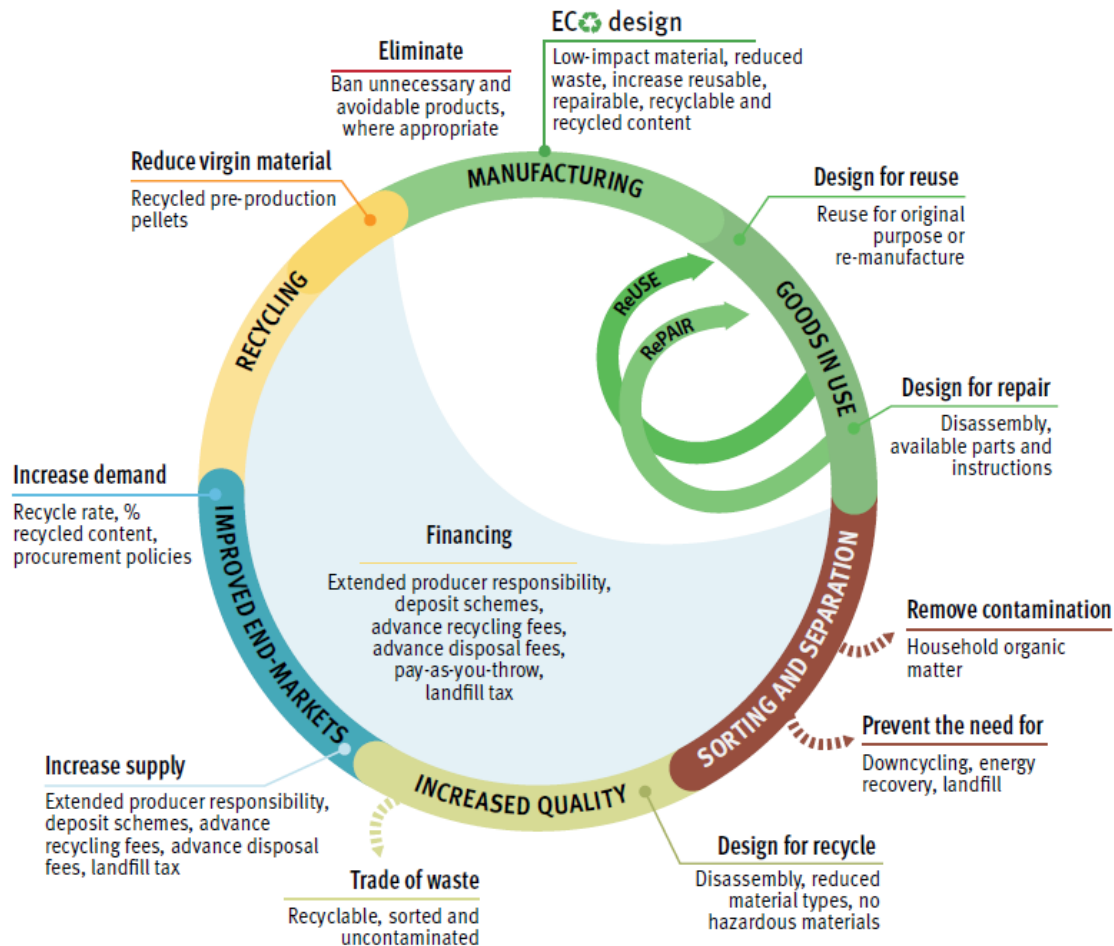
RECYCLED-VIRGIN PET PRICE RELATIONSHIP HAS CHANGED IN RECENT YEARS



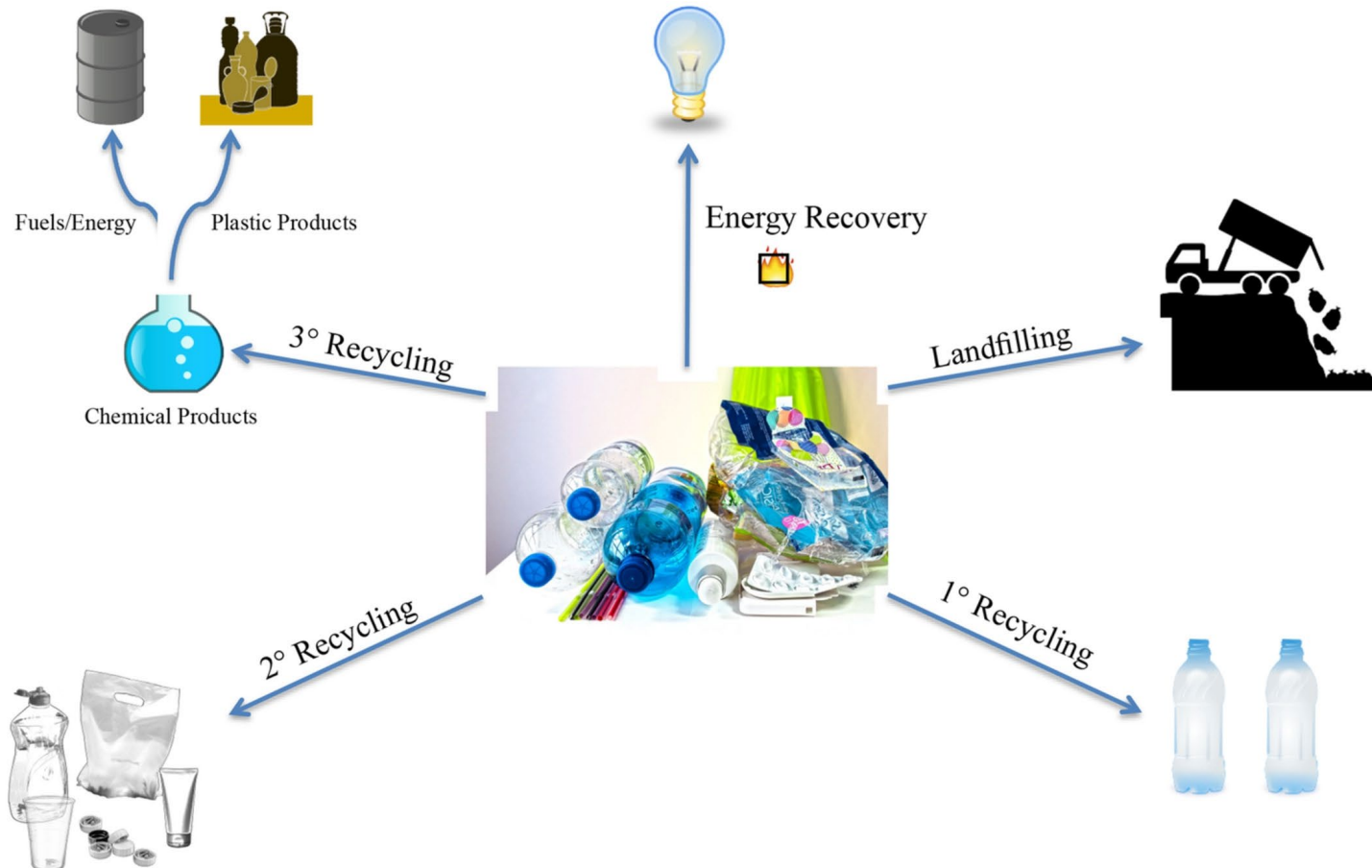
Source: S&P Global Platts

Circular economy of plastics

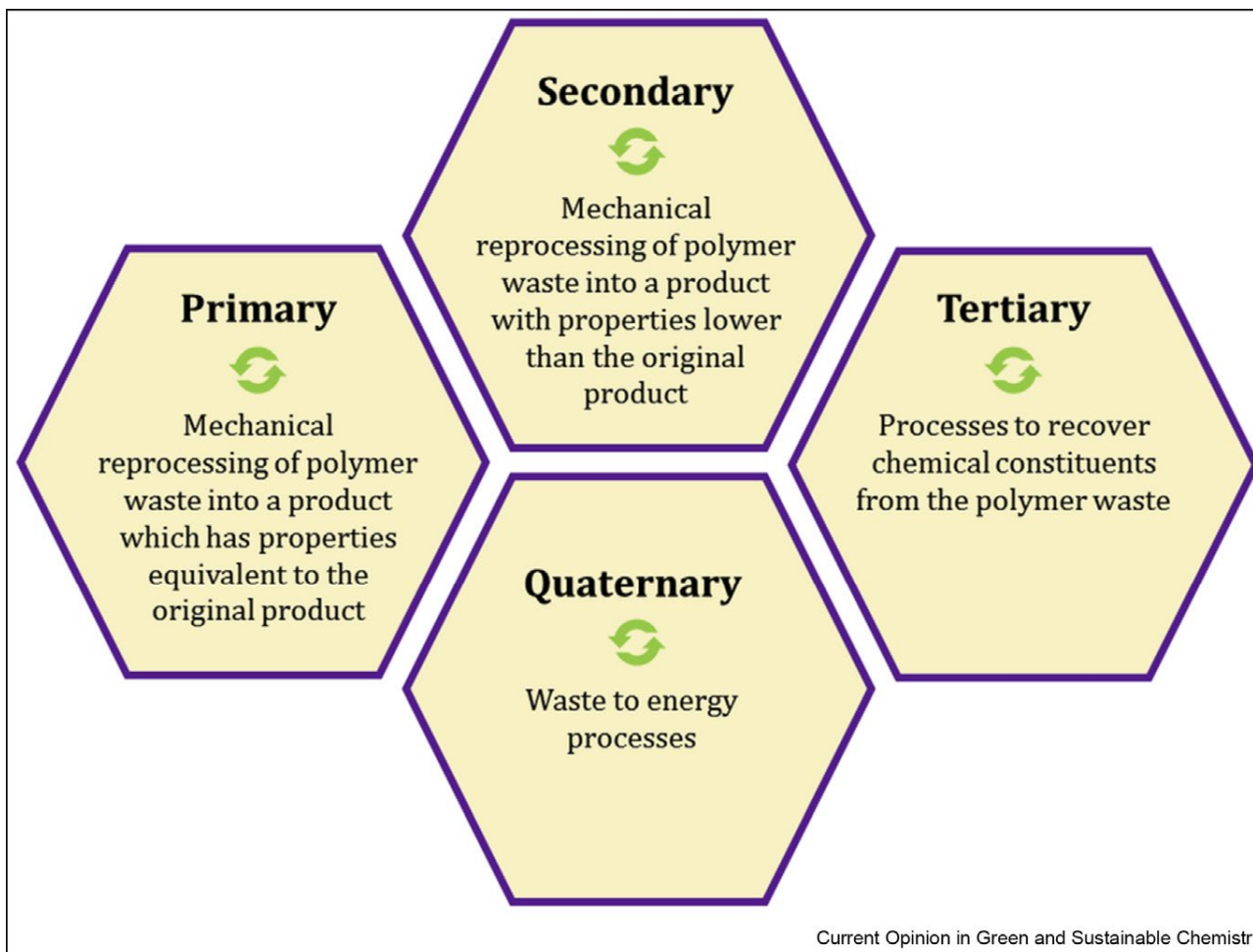
National policies towards circularity



Common fates for current plastic waste



Recycling options



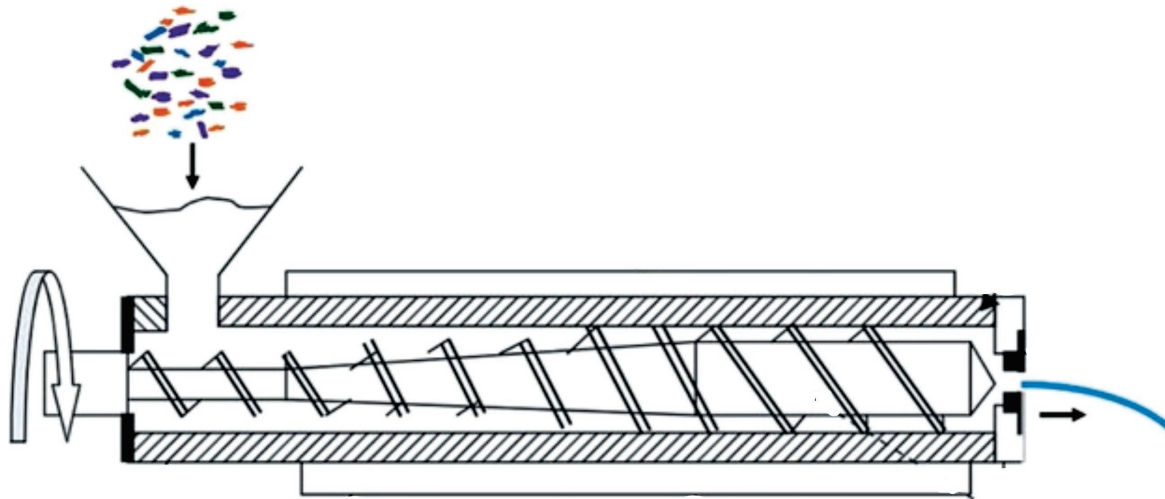
Primary Recycling

The first opportunity to practice recycling is at the processing facility, where no separation of the material is required: **considered a form of mechanical recycling.**

All major operations such extrusion, injection molding, calendaring, thermoforming, produces scraps.

Scraps can originate from rejected parts, sprues and runners generated during injection molding. In some cases, the scrap can represent 50% of the processed material (e.g.; thermoforming parts).

This scrap (or regrind) can be ground and combined with virgin resin to not only improve the economics of the process but prevent disposing of the material in landfills.



Mechanical or secondary recycling

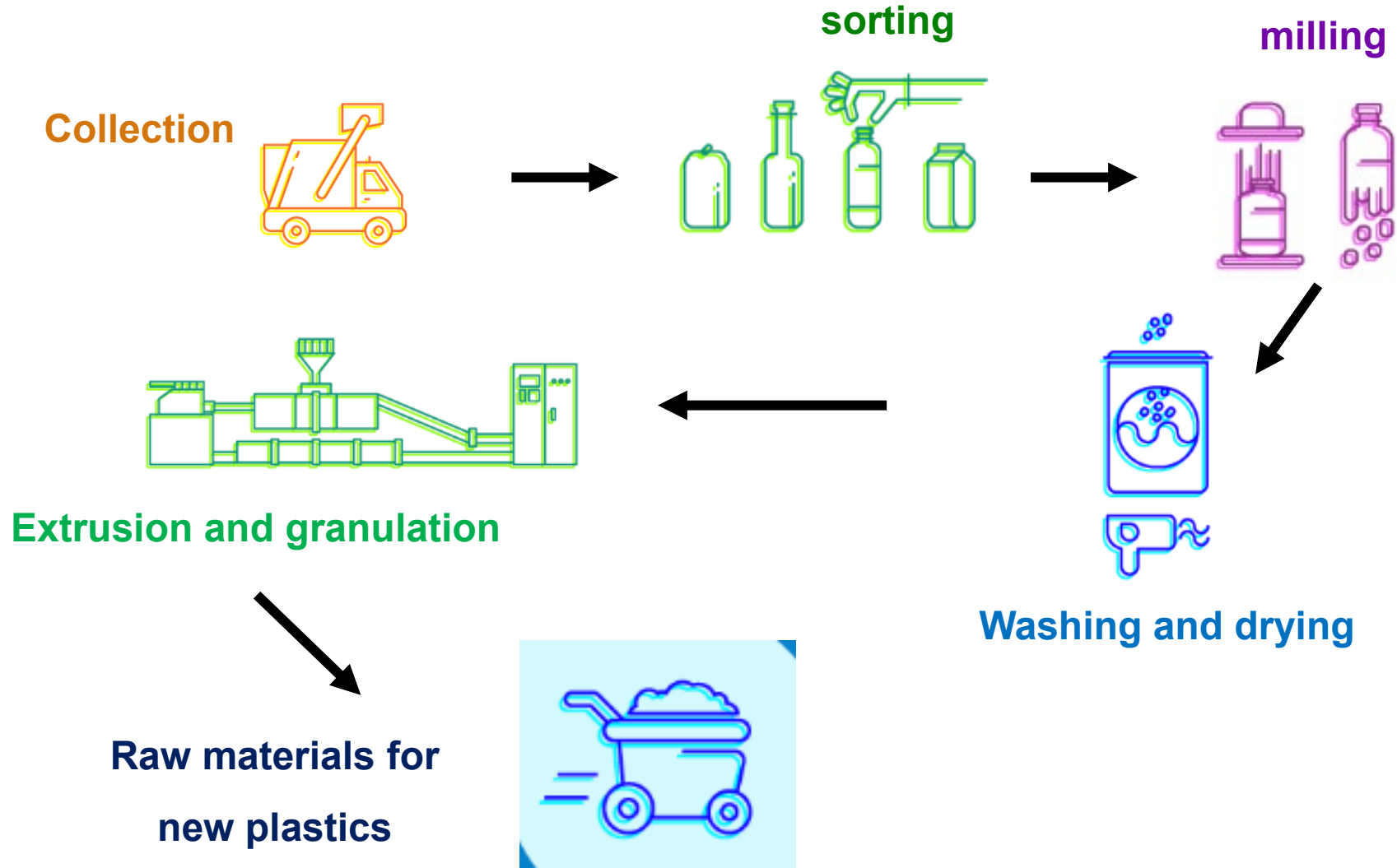
Secondary recycling refers to the use of plastics unsuitable for primary recycling using standard plastics processing equipment.

Sources of plastic wastes potentially suitable for secondary recycling processes include:

- I. post-consumer waste recovered from municipal refuse,
- II. post-consumer waste obtained from returnable packages,
- III. mixed industrial plastic waste,
- IV. industrial plastic waste consisting of a single type of plastic

Sometimes secondary recycling is also referred to as mechanical recycling, which means reusing the recovered product as material for either the original purpose or for different ones.

Mechanical or secondary recycling



Mechanical or secondary recycling

The complications arise as a result of the **mixing of polymer types and the presence of impurities** (metal, sand, glass, paper, etc.). The problem with mixed polymer types is that polymers are highly incompatible, and their properties are significantly lower than those of the pure components (*thermodynamics of mixing*). In some cases, such as in the generation of parts with thick walls (e.g., plastic lumber), mixed polymers can be tolerated.



Not only a wide variety of plastics is present in the MSW, but a wide variety of other materials are present as plastics make up only 6.2% (see table below).

Plastic Type (SPI #) ^a	Percent by Weight in MSW	Percent by Weight of Plastics
PS (6) foam	0.5	8.0
PET (1) soda bottles	1.0	16.0
HDPE (2) milk/water bottles	0.7	11.0
Bags (4) LDPE film	1.8	29.0
Other (LDPE (4), PVC (3) bottles, PS (6) caps, and other (7) plastics)	2.2	36.0
Total	6.2	100.0

Plastic in Municipal Solid Waste (MSW) of Hamilton, Ohio

The incompatibility of polymers and differences in their melting and processing temperatures, it is difficult to directly reprocess a polymeric mixture.

Technical approaches to secondary recycling include direct reprocessing, reprocessing using specialized equipment, chemical modification of mixed plastic waste (compatibilizers), using recycled in the core, using waste as a filler, and using waste as a binder for a low-cost filler.

Mechanical or secondary recycling

Barriers/challenges:

- The purity of the materials is not known;
- The contamination of the matrix with other polymers reduces the mechanical properties. Most polymers are not compatible with each other. Examples are the presence of PET impurities in a PVC matrix.

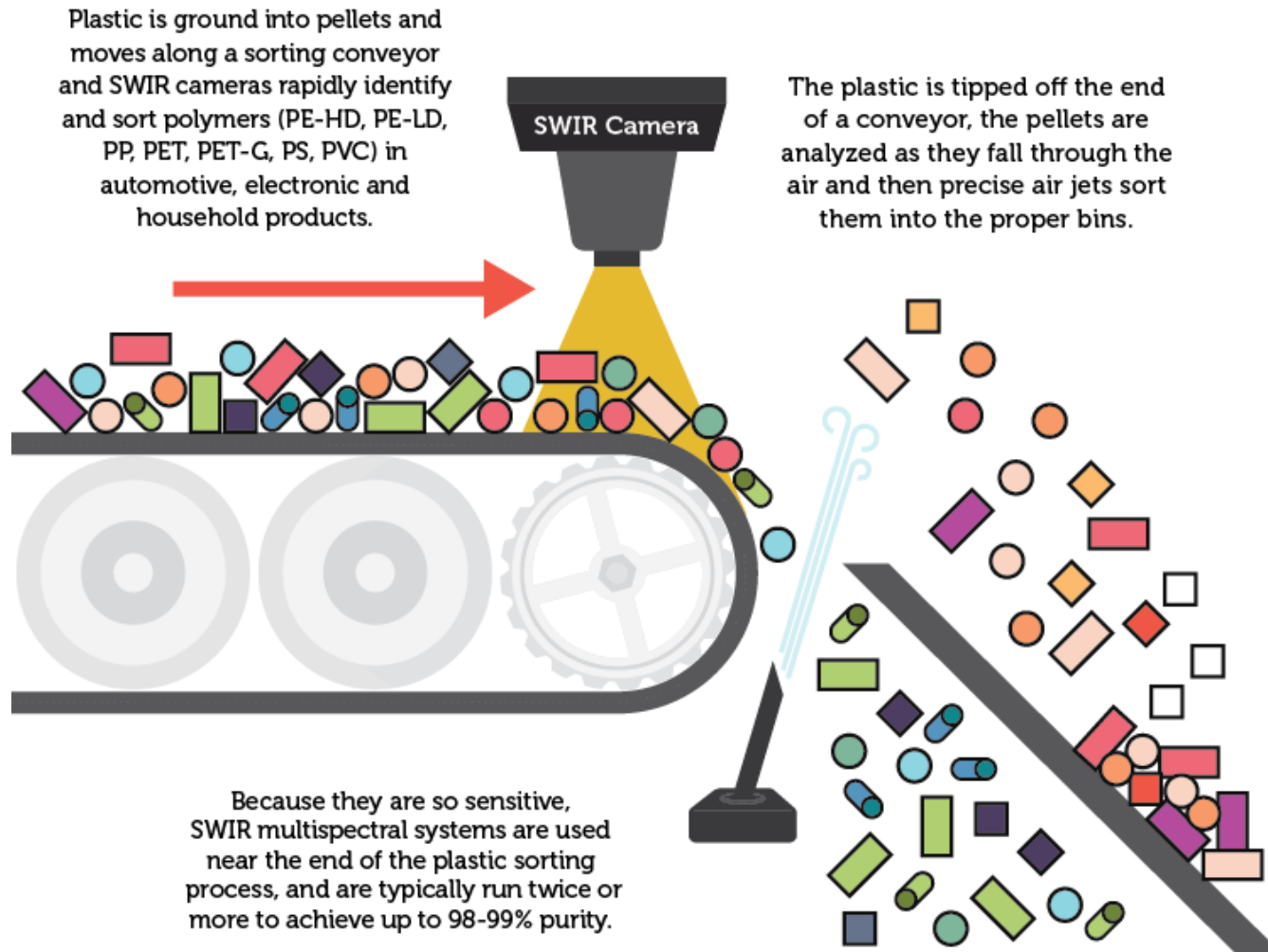


Separation techniques:

- Fourier-transform and near-infrared spectroscopy
- X-ray detection
- Optical color recognition cameras to separate clear and colored materials from each other.

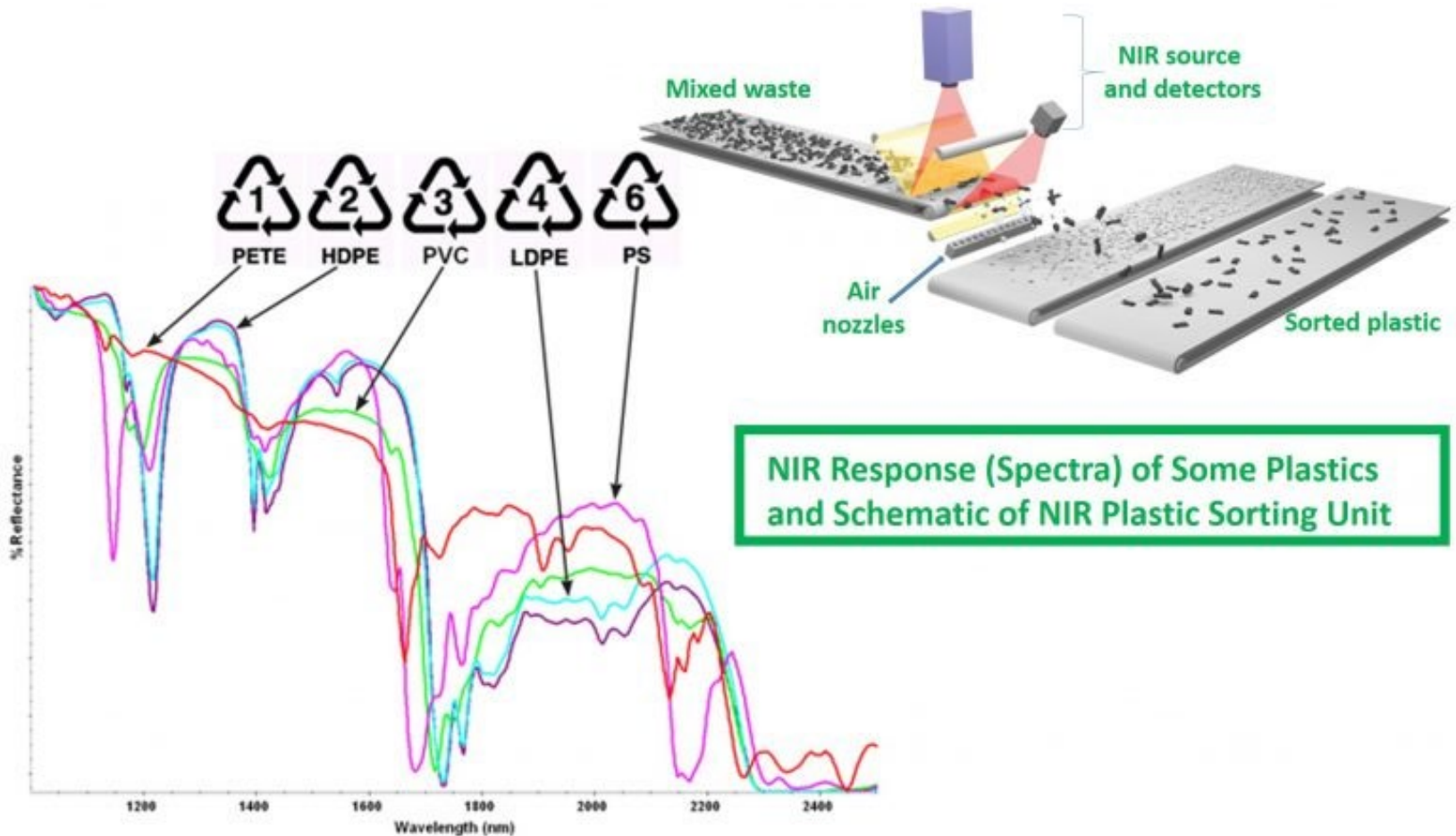
Mechanical or secondary recycling

Fourier-transform and near-infrared spectroscopy



Mechanical or secondary recycling

Fourier-transform and near-infrared spectroscopy

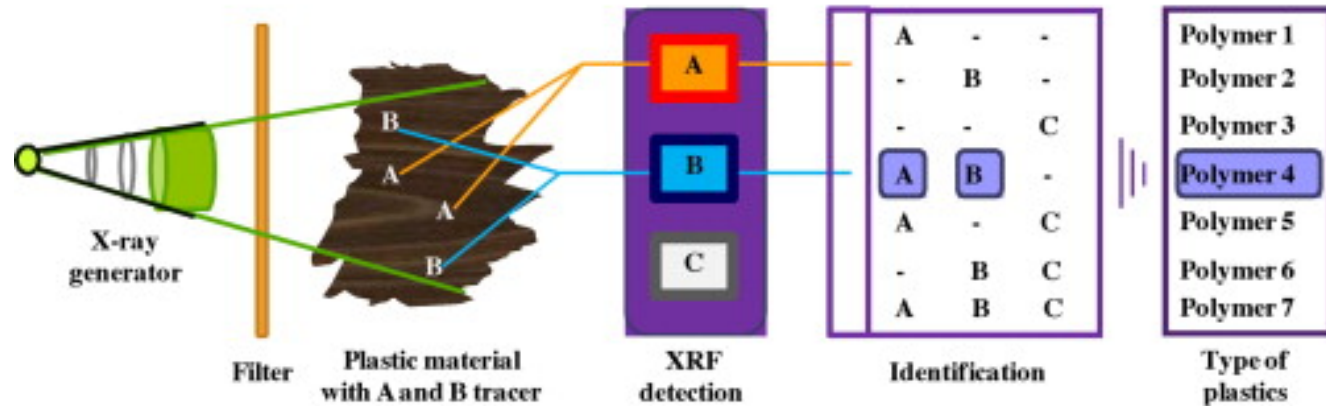


Mechanical or secondary recycling

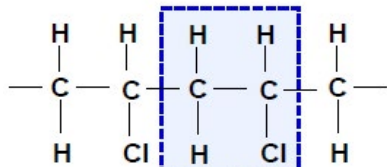


Mechanical or secondary recycling

X-ray fluorescence



PVC



Unidade mero



Tertiary of feedstock recycling

Chemical recycling

In contrast to mechanical recycling, chemical recycling offers the potential for making high-quality polymers from waste — termed ‘upcycling’. Plastic products are depolymerized into their monomeric subunits, which can then be repolymerized through controlled polymerization mechanisms into polymers of desired quality

Solvolysis

In solvolysis, polymers with cleavable groups along their backbone, such as ester bonds in PET and PLA, can be subjected to solvent-based depolymerization processes such as hydrolysis, glycolysis or methanolysis.

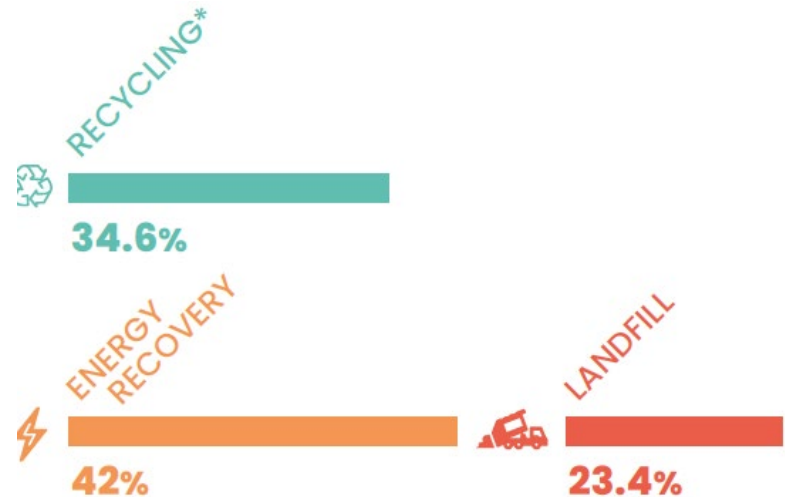
Thermolysis

In thermolysis, typically polyolefins, which do not possess hydrolysable functional groups, are pyrolyzed at temperatures of ~200–800 °C (depending on the polymer and catalyst used) in the total or partial absence of O₂. Under these conditions, the C–C bonds break, converting the polymer back into feedstock in the form of hydrocarbon oil or gas, or directly into olefin monomers.

Quaternary recycling

Incineration as a method to recover energy may be also classified as a form of recycling. Incineration (or quaternary recycling) still remains a very popular method for waste volume reduction and for energy recuperation.

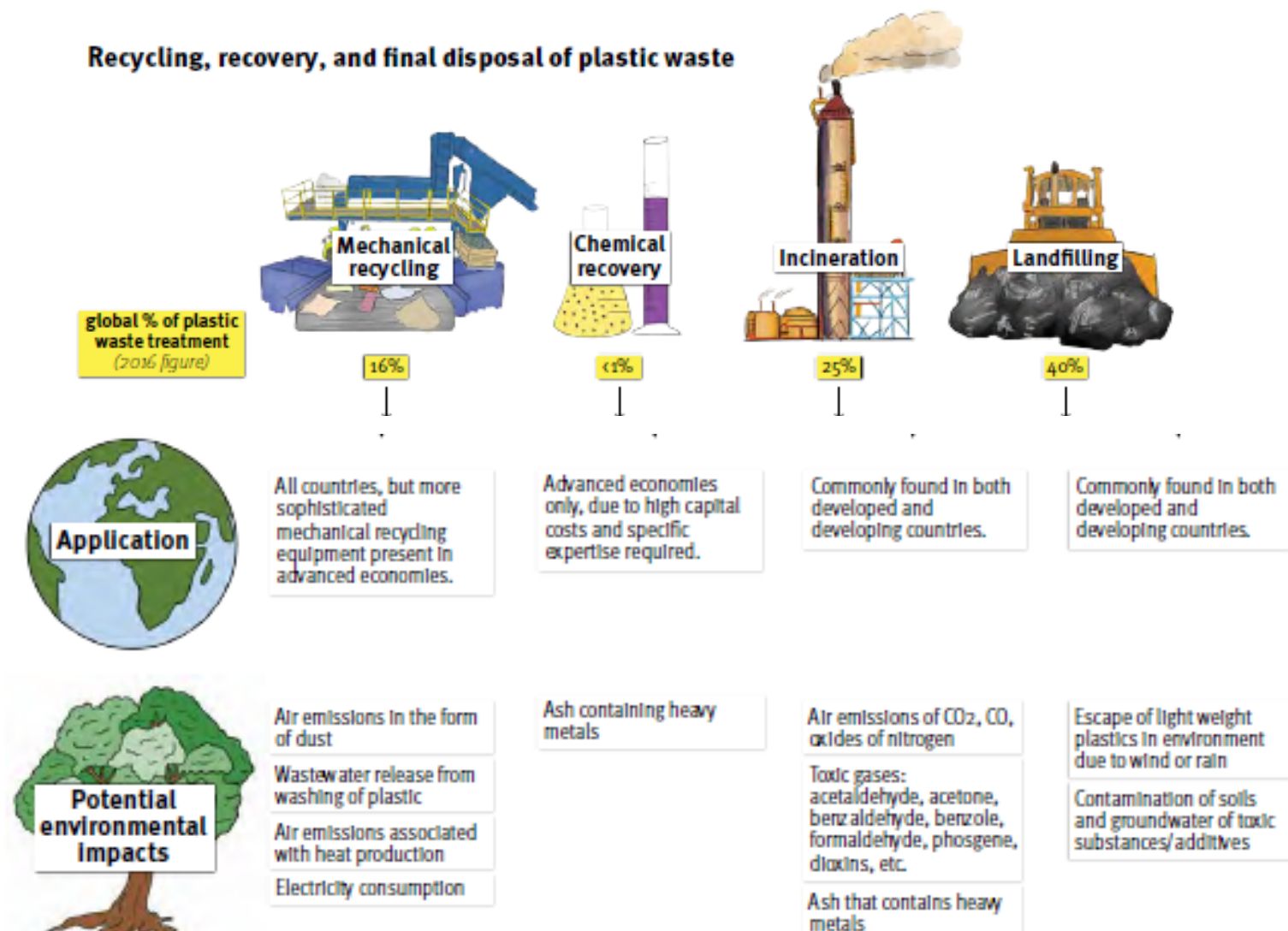
In Europe, it is the most common method of utilizing discarded plastic.



This method is especially used for processing of mixed and heavily contaminated wastes, which cannot be easily and/or economically recycled by any other method. Burning of energy-dense waste can create heat, electricity, or other forms of energy, which can be directly used in technological processes or for heating of buildings.

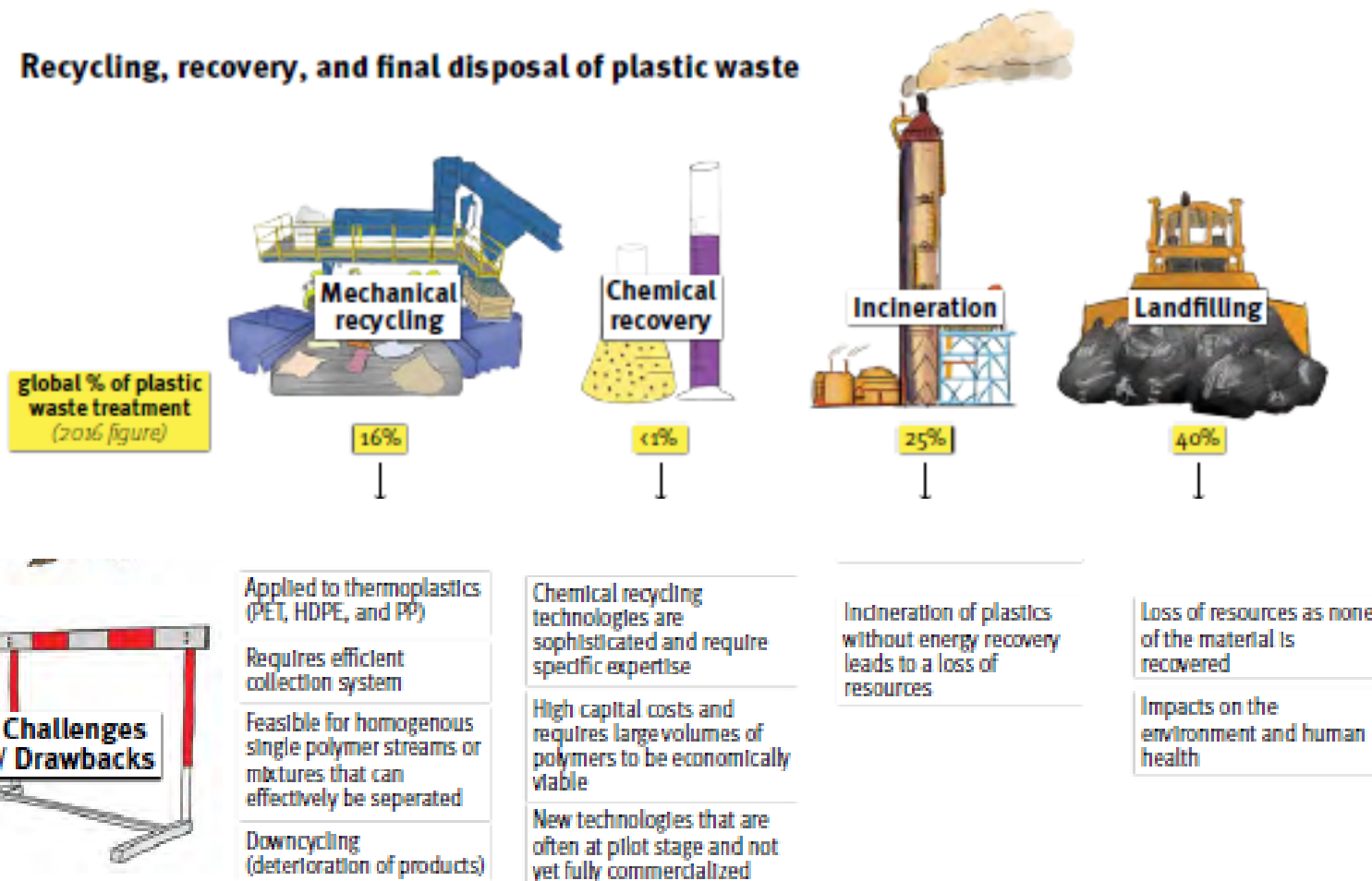
Recycling options

Recycling, recovery, and final disposal of plastic waste

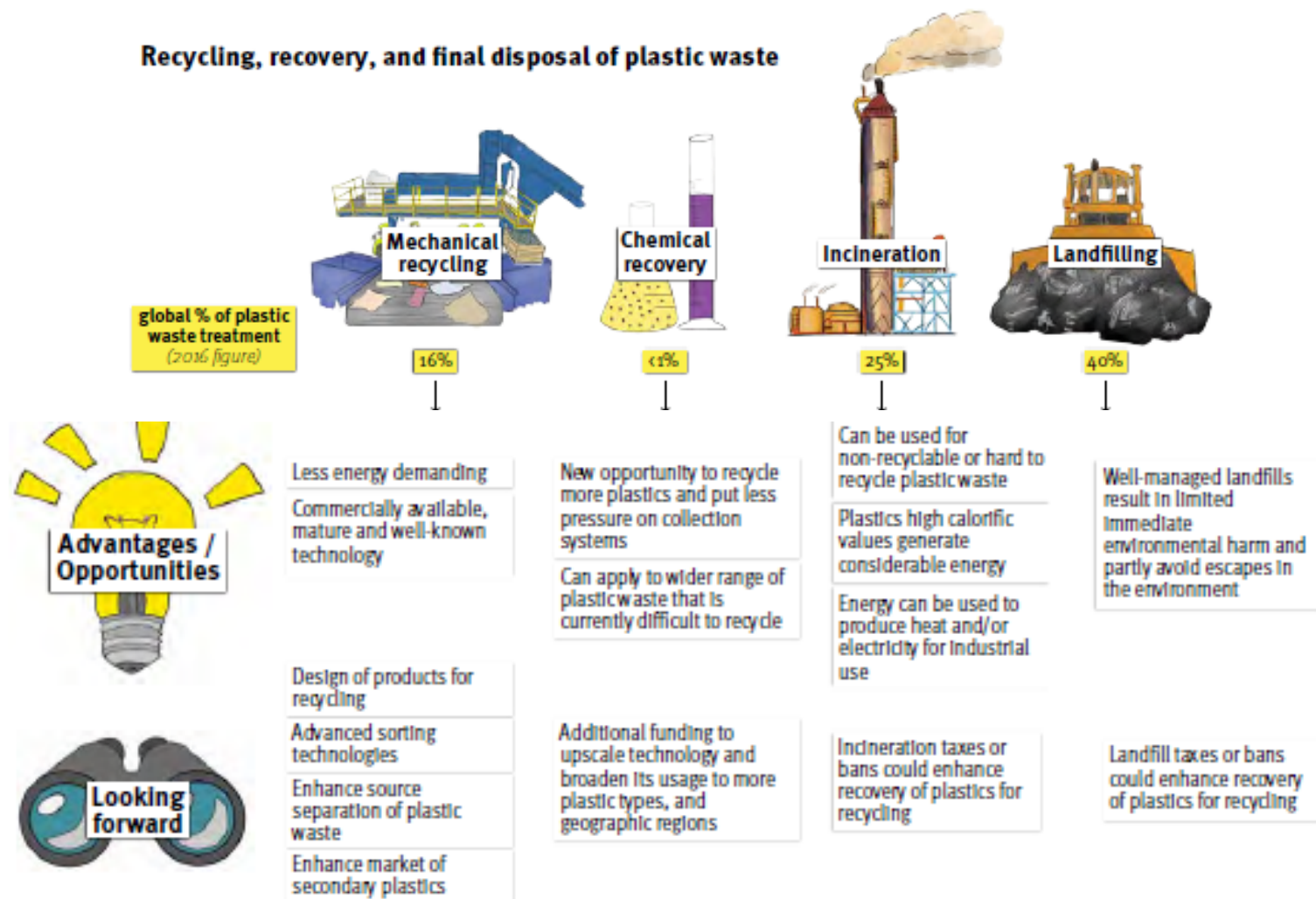


Recycling options: summary

Recycling, recovery, and final disposal of plastic waste

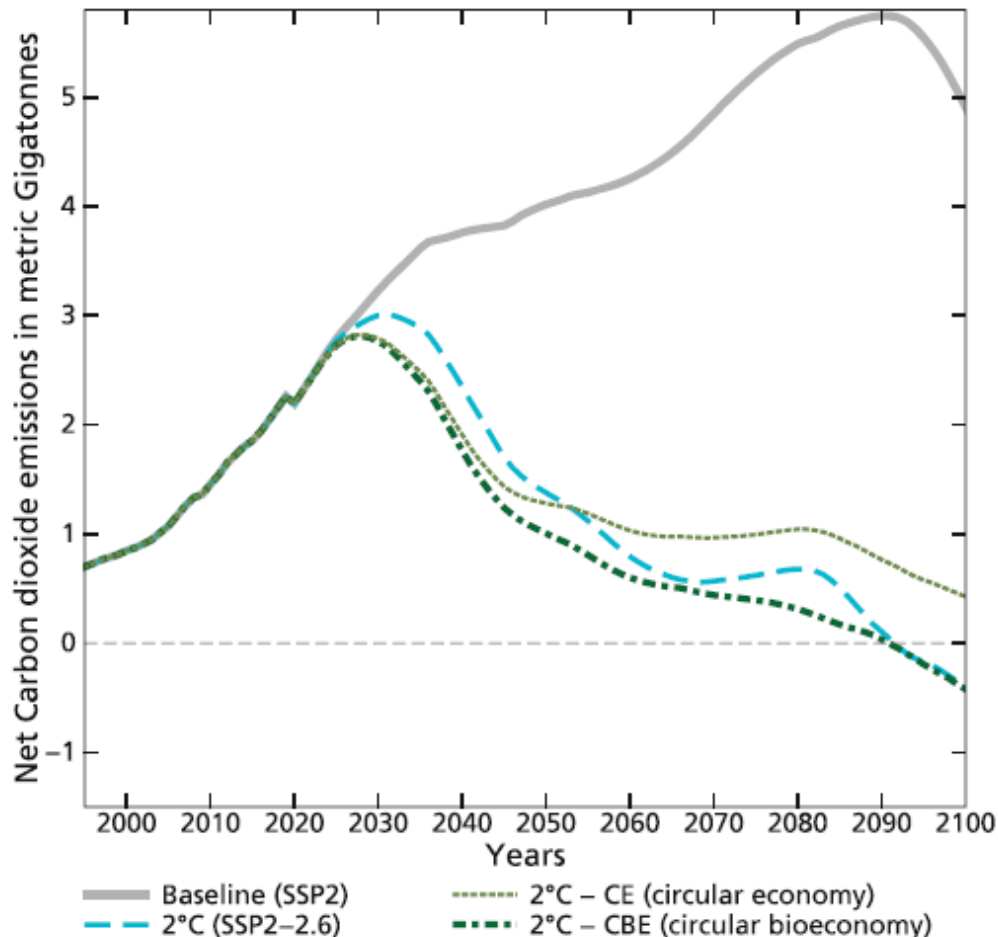


Recycling options : summary



Global carbon balance of the plastics sector over the entire life cycle

Projections under different scenarios



SSP2: middle-of-the-road socioeconomic path that results in a continued focus on fossil resources and only a small uptake of biomass as feedstock.

SSP2-2.6°C: 2°C global mean temperature change target by 2100.

CE & CBE: a global paradigm shift toward a CE involving all relevant actors is assumed. Next to gradually phasing out landfilling, the authors assume that policies incentivizing circular product design, standardized plastic types and avoidance of additives, opaque colours and multimaterial plastic products will increase sorting and recycling efficiencies.

Extended Data Fig. 3 | Comparing the net CO₂ emissions of the four scenarios for the global plastic sector. These emission lines are the same as the solid net emission lines of Fig. 3; biogenic emissions are assumed to be renewable and therefore have no net contribution to climate change.

Plastics are the most complicated material family. It will be very difficult to make a 50% cut in CO₂ emissions while predicted demand doubles. Here are some suggestions:

- ❑ Reduce the variety of plastics in use to simplify recycling and increase recovery rates.
- ❑ Replace all possible disposable packaging with long life packaging in continuous re-use and extend the life of all non-disposable plastic goods.
- ❑ Promote life extension for other products, including vehicles, which contain many plastic components, as part of a general strategy for reducing demand for new materials (economic impact?)
- ❑ Development of bio-based plastics.